Stat 266 – Unit 6 – Video Set 2 – Class Examples

Summary:

Method of Transformations (Univariate)

Note: We will only consider continuous random variables for this method.

Consider $U = h(Y)$ to be a function of the continuous random variable Y, with density $f_Y(y)$.

If $U = h(Y)$ is either increasing or decreasing (one-to-one) for all y (where $f_Y(y) > 0$) then $U = h(Y)$ has density

important.

$$f_U(u) = f_Y\left(h^{-1}(u)\right)\left|\frac{d}{du}h^{-1}(u)\right|.$$

why this is true?

Note: If $u = h(y)$ then $y = h^{-1}(u)$ is the inverse function.

Following the method of distribution.

If $h(u)$ is increasing

$$\underbrace{P(U \le u)}_{F_U(u)} = P(h(Y) \le u) = P(Y \le \underbrace{h^{-1}(u)}_{F_Y(h^{-1}(u))}).$$

take derivative with respect to u

$$\frac{dF_U(u)}{du} = \frac{dF_Y(h^{-1}(u))}{du} = \underbrace{\frac{dF_Y(h^{-1}(u))}{dy}}_{f_Y(h^{-1}(u))} \cdot \frac{h^{-1}(u)}{du}.$$

If $u = h(Y)$ is decreasing.

$$\underbrace{P(U \le u)}_{F_U(u)} = P(h(Y) \le u) = P(Y \ge h^{-1}(u))$$

$$= 1 - \boxed{P(Y \le h^{-1}(u))} = 1 - F_Y(h^{-1}(u)).$$

take derivative both sides, we have

$$\frac{dF_U(u)}{du} = \frac{d(1 - F_Y(h^{-1}(u)))}{du} = -f_Y(h^{-1}(u)) \cdot \frac{dh^{-1}(u)}{du}.$$

since $\frac{d(h^{-1}(u))}{du} < 0$

so $-\frac{d(h^{-1}(u))}{du} = \left|\frac{d(h^{-1}(u))}{du}\right|$

Example A:

The amount of time waiting to be served, Y, is described by the following density

$$f(y) = \begin{cases} \frac{1}{\theta} e^{-y/\theta} & , \quad y > 0 \\ 0 & , \quad else \end{cases}.$$

a) What is the probability distribution of $U = aY + b$, where $a > 0$?

since $u = aY + b$ $\quad Y = \frac{u-b}{a} = h^{-1}(u)$.

$$f_U(u) = \frac{1}{\theta} e^{-\frac{h^{-1}(u)}{\theta}} \cdot \frac{dh^{-1}(u)}{du} = \frac{1}{\theta} e^{-\frac{u-b}{a\theta}} \cdot \frac{1}{a}$$

$$= \frac{1}{a\theta} e^{-\frac{u-b}{a\theta}}.$$

This is only valid for $Y > 0$, that means $h^{-1}(u) > 0 \Rightarrow \frac{u-b}{a} > 0$ $\quad u > b$.

b) What is the probability distribution of $U = Y^a$, where $a > 0$?

$u = Y^a \Rightarrow Y = u^{\frac{1}{a}}$

$f_U(u) = \frac{1}{\theta} e^{-\frac{u^{\frac{1}{a}}}{\theta}} \cdot \left| \frac{1}{a} u^{\frac{1}{a}-1} \right|$

since Y must greater than 0,

$u > 0$.

$$f_U(u) = \begin{cases} \frac{1}{\theta} e^{-\frac{u^{\frac{1}{a}}}{\theta}} \frac{1}{a} u^{\frac{1}{a}-1} & u > 0. \\ 0 & \text{otherwise} \end{cases}$$

Example B:

Consider the random variable R with density

$$f(r) = \begin{cases} 4\pi r^2 e^{(-4/3)\pi r^3} & , r > 0 \\ 0 & , else \end{cases}$$

What is the variance of R^3?

$Y = R^3$ $\quad r = \sqrt[3]{Y}$. $\quad f_R(x) = 4\pi (\sqrt[3]{y})^2 e^{(-\frac{4}{3})\pi (\sqrt[3]{y})^3}$

$$\frac{dR}{dy} = \frac{1}{3} \cdot y^{-\frac{2}{3}} \cdot \qquad \cdot \frac{dR}{dy}$$

$$= 4\pi (\sqrt[3]{y})^2 e^{(-\frac{4}{3})\pi y} \cdot \frac{1}{3} y^{-\frac{2}{3}}$$

$$= \frac{4\pi}{3} e^{-\frac{4}{3}\pi y}.$$

This is an Exponential distribution with $\theta = \frac{3}{4\pi}$

$E(R^3) = \frac{1}{\theta} = \frac{4\pi}{3}$ $\quad Var(R^3) = \frac{1}{\theta^2} = (\frac{4\pi}{3})^2$

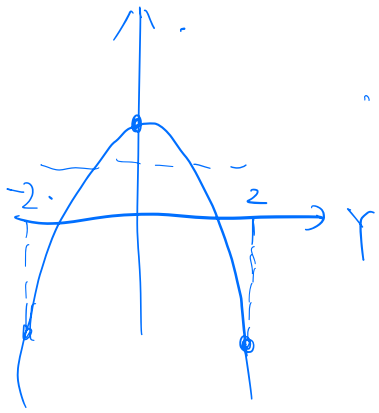
Example C: If Y is a continuous random variable with density function

$$f_Y(y) = \begin{cases} \dfrac{3y^2}{16} & , \quad -2 < y < 2 \\ 0 & , \quad else \end{cases},$$

what is the density function of $U = 1 - 6Y^2$?

4

$U \in [-23, 1]$.



$P(U \le u) = P(1 - 6Y^2 \le u)$.

$\frac{1-u}{6} \le Y^2 \Rightarrow \begin{cases} Y \le -\sqrt{\frac{1-u}{6}} \\ Y > \sqrt{\frac{1-u}{6}} \end{cases}$.

$= P\left(Y \le -\sqrt{\frac{1-u}{6}}\right) + P\left(Y \ge \sqrt{\frac{1-u}{6}}\right)$

$P(U \le u) = P\left(Y \le -\sqrt{\frac{1-u}{6}}\right) + 1 - P\left(Y \le \sqrt{\frac{1-u}{6}}\right)$.

take derivative $f_U(u) = \dfrac{d\left(F_Y\left(-\sqrt{\frac{1-u}{6}}\right)\right)}{du} - \dfrac{d\left(F_Y\left(\sqrt{\frac{1-u}{6}}\right)\right)}{du}$

$= \dfrac{3\left(\sqrt{\frac{1-u}{6}}\right)^2}{16} \cdot \dfrac{d\left(-\sqrt{\frac{1-u}{6}}\right)}{du}$

$- \dfrac{3\left(\sqrt{\frac{1-u}{6}}\right)^2}{16} \dfrac{d\left(\sqrt{\frac{1-u}{6}}\right)}{du}$.

$= \dfrac{3\left(\sqrt{\frac{1-u}{6}}\right)^2}{16} \cdot \frac{1}{2} \cdot \left(\frac{1-u}{6}\right)^{-\frac{1}{2}} \cdot \frac{1}{6} \times 2$.

Example D : The density of Y is given as

$$f(y)=\begin{cases}(3/2)y^2 \ , & -1<y<1 \\ 0 \ , & else\end{cases}$$

For a constant $a>0$,

a) What is the probability distribution of $U=aY$?

b) What is the probability distribution of $V=a-Y$?

c) What is the probability distribution of $W=Y^2$?

a). $Y=\frac{U}{a}$ $\quad f_U(u)=\begin{cases}\frac{3}{2}\cdot\left(\frac{u}{a}\right)^2\cdot\frac{1}{a} & -a<u<a \\ 0 & \text{otherwise.}\end{cases}$

b) $Y=a-V$. $\quad f_V(v)=\begin{cases}\frac{3}{2}(a-v)^2\cdot(-1) & a-1<u<a+1 \\ 0 & \text{otherwise.}\end{cases}$

c). $W=Y^2$. Not increasing or decreasing. $\quad \underset{\underset{F_W(w)}{\parallel}}{P(W\le w)}=P(Y^2\le w)$

$$= P(-\sqrt{w} \le Y \le \sqrt{w}) = \int_{-\sqrt{w}}^{\sqrt{w}} \frac{3}{2} y^2 dy$$

Take derivative both sides

$$\frac{dF_W(w)}{dw} = f_W(w) = \frac{d \int_{-\sqrt{w}}^{\sqrt{w}} \frac{3}{2} y^2 dy}{dw}$$

$$= \frac{d \int_0^{\sqrt{w}} \frac{3}{2} y^2 dy + \int_{-\sqrt{w}}^{0} \frac{3}{2} y^2 dy}{dw}$$

$$= \frac{d \int_0^{\sqrt{w}} \frac{3}{2} y^2 dy - \int_0^{-\sqrt{w}} \frac{3}{2} y^2 dy}{dw} = \frac{3}{2} w \cdot \frac{1}{2} \frac{1}{\sqrt{w}} - \frac{3}{2} w$$

$$-\frac{1}{2\sqrt{w}} = \frac{1}{\sqrt{w}} \cdot \left(\frac{3}{2}\right)$$

$$= \frac{3}{2}\sqrt{w} \quad \text{and } w < 1.$$

Example E: If Y is a continuous random variable with density function

$$f_Y(y) = \begin{cases} \dfrac{y^2}{3} & , \quad -1 < y < 2 \\ 0 & , \quad else \end{cases},$$

what is the density function of $U = Y^2$?

$u \in [0, 4)$.

$$P(U \le u) = P(Y^2 \le u)$$

$$= P(-\sqrt{u} \le Y \le \sqrt{u})$$

Now consider $y \in (-1, 2)$.

So when $-1 < -\sqrt{u} \Rightarrow \sqrt{u} < 1 \Rightarrow u < 1$

$$P(-\sqrt{u} \le Y \le \sqrt{u}) = \int_{-\sqrt{u}}^{\sqrt{u}} \frac{y^2}{3}\, dy \quad (1)$$

when $4 \geq u > 1$ $\quad P(-\sqrt{u} \leq Y \leq \sqrt{u}) = P(-1 \leq Y \leq \sqrt{u})$

$$= \int_{-1}^{\sqrt{u}} \frac{y^2}{3}\, dy \quad ②$$

Take deriviative of ①, we have

$$\frac{u}{3} \cdot \frac{1}{2\sqrt{u}} - \frac{u}{3}\left(-\frac{1}{2\sqrt{u}}\right) = \frac{\sqrt{u}}{3} \qquad 0 < u < 1.$$

Take deriviative of ②. we have

$$\frac{u}{3} \cdot \frac{1}{2\sqrt{u}} = \frac{\sqrt{u}}{6} \qquad u \geq 1.$$

$$f_U(u) = \begin{cases} \dfrac{\sqrt{u}}{3} & 0 < u < 1 \\ \dfrac{\sqrt{u}}{6} & 4 \geq u \geq 1 \\ 0 & \text{otherwise.} \end{cases}$$